

FE-CH-USSD-01 - USSD Channel

Version: v1.0 **Bundle:** 99_frontend **Status:** Draft for review **Primary owner:** Channels / integration team **Related DDs:** FE-APP-04, DD_ILM-CFG-01, DD_SUB-LM-01, DD_BIL-01, DD_BIL-04, DD_PAY-GW-01, DD_TCK-01, DD_NOT-01, DD_EM-CFG-03, DD_EM-CFG-04, DD_CUST-INT-01

MVP simplification baseline: first release USSD scope is limited to language selection, customer lookup, balance/debt status, payment instructions/status, support ticket creation/status, and callback request. Do not implement package changes, subscription migration, complex Customer 360 views, or local business state in USSD v1.

1. Purpose

FE-CH-USSD-01 defines the USSD self-care channel for customers using feature phones or low-bandwidth mobile access.

The USSD channel should support simple customer self-service such as account lookup, balance, invoice/debt status, package/subscription summary, payment instructions, ticket creation/status, language selection, and callback request.

USSD is a channel adapter. It does not own customer, billing, payment, subscription, ticket, or notification logic.

2. Scope And Boundaries

Capability	Owner
USSD session, menu, screen rendering, telco gateway adapter	FE-CH-USSD-01
Customer/account/contact lookup	ILM-CFG-01
Subscription summary/status	SUB-LM-01
Balance, invoice, payment status	BIL-01/BIL-02/BIL-04
Payment initiation/instructions	PAY-GW-01 and BIL payment APIs
Support ticket create/status	TCK-01
Customer notifications	NOT-01
Interaction history	CUST-INT-01
Permissions/channel policies	EM-CFG-03 / channel config

Boundary rules:

- USSD does not apply payments locally.
- USSD does not change subscription state locally.
- USSD does not close tickets locally.
- USSD must call owning module APIs with idempotency keys.
- USSD screens must be short, language-aware, and timeout-safe.

3. Process Overview

```
sequenceDiagram
    participant TELCO as USSD Gateway
    participant USSD as SOPHIX USSD Adapter
    participant ILM as ILM-CFG-01
    participant BIL as Billing
    participant SUB as SUB-LM
    participant TCK as TCK-01
    participant INT as CUST-INT

    TELCO->>USSD: session request MSISDN + input
    USSD->>USSD: load session and menu state
    USSD->>ILM: resolve customer by MSISDN
    USSD->>BIL: get balance/debt when selected
    USSD->>SUB: get subscription status when selected
    USSD->>TCK: create/list ticket when selected
    USSD->>INT: record interaction
    USSD-->>TELCO: CON/END text response
```

4. Supported V1 Menus

Menu	Purpose	Module APIs
Account summary	Customer name/account status.	ILM - CFG - 01.
Balance / amount due	Current debt, next due date, dunning state.	BIL - 01, BIL - 04.
Pay / payment instructions	Show paybill/till/reference or initiate STK where enabled.	PAY - GW - 01, BIL payment APIs.
Subscription status	Active/suspended/restricted package summary.	SUB - LM - 01.
Support	Create simple ticket or view latest ticket status.	TCK - 01.
Callback request	Create ticket/activity for agent callback.	TCK - 01 or CVM/Backoffice queue.
Language	Set language preference where allowed.	ILM - CFG - 01 / preferences API.

5. Data Model

5.1 ussd_session

What it is for: stores an active USSD session and menu state.

When to use it: create/update on every gateway request until session ends or times out.

Sample meaning: ussd-sess-001 is customer phone 254700000001 currently viewing balance menu.

Column	Type	Purpose	Sample
session_id	TEXT PK	Internal session id.	ussd-sess-001
operator_code	TEXT	Tenant/operator.	WIK
gateway_session_id	TEXT	Telco gateway session id.	gw-abc-001
msisdn	TEXT	Customer phone number, normalized.	254700000001
customer_id	TEXT NULL	Resolved customer.	CUS-100045
account_id	TEXT NULL	Resolved account.	ACC-900045
language_code	TEXT	Current language.	en
current_menu_code	TEXT	Menu state.	BALANCE
session_data_json	JSONB	Temporary menu/session data.	{"selectedAccount": "ACC-900045"}
status	TEXT	ACTIVE, ENDED, TIMED_OUT, FAILED.	ACTIVE
started_at	TIMESTAMP	Start time.	2026-08-30T08:00:00Z
last_seen_at	TIMESTAMP	Last request time.	2026-08-30T08:01:00Z
ended_at	TIMESTAMP NULL	End time.	null

Recommended indexes:

- unique (operator_code, gateway_session_id).
- (msisdn, status).
- (last_seen_at, status).

5.2 ussd_menu_definition

What it is for: stores configurable menu definitions per operator/language.

When to use it: seed menus for Kenya/WIK and override labels/options for other operators.

Sample meaning: menu MAIN in English shows balance, payment, subscription, support, and language options.

Column	Type	Purpose	Sample
menu_code	TEXT	Menu id.	MAIN
operator_code	TEXT	Tenant/operator.	WIK

Column	Type	Purpose	Sample
language_code	TEXT	Language.	en
menu_text	TEXT	Screen text template.	1 Balance\n2 Pay\n3 Support
options_json	JSONB	Option routing.	{"1":"BALANCE","2":"PAY","3":"SUPPORT"}
requires_customer	BOOLEAN	Whether customer resolution is required.	true
enabled	BOOLEAN	Menu enabled.	true

Recommended indexes:

- unique (operator_code, language_code, menu_code).

5.3 ussd_request_log

What it is for: stores request/response trace for operational support and dispute handling.

When to use it: create one row per gateway request.

Sample meaning: usrl-001 stores an inbound 1 selection and the balance screen returned.

Column	Type	Purpose	Sample
request_log_id	TEXT PK	Log id.	usrl-001
session_id	TEXT	USSD session.	ussd-sess-001
operator_code	TEXT	Tenant/operator.	WIK
gateway_request_id	TEXT NULL	Gateway message id.	gw-msg-001
input_text	TEXT NULL	Customer input.	1
response_type	TEXT	CON, END.	CON
response_text	TEXT	Rendered response.	Your balance is KES 1200...
error_code	TEXT NULL	Error if failed.	null
created_at	TIMESTAMP	Request time.	2026-08-30T08:01:00Z

Recommended indexes:

- (session_id, created_at).

- (gateway_request_id).

6. Gateway API

The exact request format depends on the USSD aggregator. The internal normalized endpoint should be:

```
POST /api/channels/ussd/sessions
Content-Type: application/json

{
  "operatorCode": "WIK",
  "gateway": "SAFARICOM_AGGREGATOR",
  "gatewaySessionId": "gw-abc-001",
  "gatewayRequestId": "gw-msg-001",
  "msisdn": "254700000001",
  "inputText": "1",
  "shortCode": "*123#",
  "requestedAt": "2026-08-30T08:01:00Z"
}
```

Response:

```
{
  "responseType": "CON",
  "text": "Balance: KES 1200\n1 Pay now\n2 Main menu"
}
```

7. Module API Usage

USSD action	Owning module	API
Resolve customer by phone	ILM-CFG-01	GET /api/customers/lookup?msisdn=...
Get account summary	ILM-CFG-01	GET /api/customers/{customer_id}/summary
Get subscription summary	SUB-LM-01	GET /api/subscriptions?customerId=...
Get balance/debt	BIL	GET /api/accounts/{account_id}/balance
Get dunning status	BIL-04	GET /api/dunning-states?accountId=...
Payment instructions	PAY-GW-01	POST /api/payments/instructions
Create ticket	TCK-01	POST /api/tickets
Latest ticket status	TCK-01	GET /api/tickets?customerId=...&limit=1
Record interaction	CUST-INT-01	POST /api/customer-interactions

8. Security And Privacy

- Use MSISDN-based lookup carefully; do not expose sensitive data before customer match.
- For risky actions, require PIN/OTP or route to assisted support.
- Do not show full ID/KYC values.
- Keep screens short and avoid sensitive details in request logs where possible.
- Apply rate limits by MSISDN, gateway, and short code.
- Mask MSISDN in logs outside operational support views.

9. Configuration

Config	Purpose
shortCode	Operator USSD code.
defaultLanguage	Default menu language.
enabledMenus	Enabled menu codes.
sessionTimeoutSeconds	Gateway/session timeout.
maxMenuDepth	Prevent long sessions.
paymentInstructionMode	Paybill/reference, STK push, or both.
supportTicketCategoryMap	USSD option to TCK ticket type.
customerLookupPolicy	Single/multiple/no account behavior.

10. Test Scenarios

1. New session resolves customer by MSISDN.
2. Unknown MSISDN shows safe registration/support message.
3. Balance menu calls billing API and renders short response.
4. Payment instruction menu calls PAY-GW/BIL API.
5. Support menu creates TCK ticket with idempotency key.
6. Duplicate gateway request returns same response.
7. Timeout marks session TIMED_OUT.
8. Interaction is written to CUST-INT-01.